# The Ly49Q receptor plays a crucial role in neutrophil polarization and migration by regulating raft trafficking.

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## Evidence abstract

Neutrophils rapidly undergo polarization and directional movement to infiltrate the sites of infection and inflammation.

Here, we show that an inhibitory MHC I receptor, Ly49Q, was crucial for the swift polarization of and tissue infiltration by neutrophils.

During the steady state, Ly49Q inhibited neutrophil adhesion by preventing focal-complex formation, likely by inhibiting Src and PI3 kinases.

However, in the presence of inflammatory stimuli, Ly49Q mediated rapid neutrophil polarization and tissue infiltration in an ITIM-domain-dependent manner.

These opposite functions appeared to be mediated by distinct use of effector phosphatase SHP-1 and SHP-2.

Ly49Q-dependent polarization and migration were affected by Ly49Q regulation of membrane raft functions.

We propose that Ly49Q is pivotal in switching neutrophils to their polarized morphology and rapid migration upon inflammation, through its spatiotemporal regulation of membrane rafts and raft-associated signaling molecules.

## Document metadata

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## Evidence checklist

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## Source continuity

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